

INDIAN ECONOMY

LEVEL
II



LECTURES

- NANO UREA
- PRADHANMANTRI BHARTIYA JANURVARAK PARIYOJNA (PMBJP)

MODULE - 60





NANO UREA ... TOWARDS PRECISION AND SUSTAINABLE AGRICULTURE!

- IFFCO developed Nano technology based Nano-Urea (liquid) fertiliser. It is environment friendly.
- It addresses the imbalanced and excessive use of conventional urea.
- This Nano fertiliser has been developed indigenously for the first time in the World at IFFCO – Nano-Biotechnology Research Centre (NBRC), Kalol, Gujarat.
- It is through a proprietary patented technology.
- Nano Urea (liquid) is a source of Nitrogen, which is a major essential nutrient required for proper growth and development of a plant.



Nano Urea does not involve any government subsidy. It will be made available to farmers at a 10% lower price than a bag of subsidized urea.



A BRIEF ON NANO UREA

- Nano urea contains nitrogen particles of 20 to 50 nanometers.
- One conventional urea prill, whose thickness is around 2.8 millimeters, is equal to 55,000 nano urea particles in size.
- 500 ML costs Rs. 240, whereas the conventional subsidized urea is sold at Rs. 266.5 per 45 kg bag.
- The Parliamentary Standing Committee of Fertilizers is of the view that transfer of nano urea technology to the private sector could be a game changer for the agriculture sector.

Farmer can save a lot on the transportation front also!

WHY IS NITROGEN IMPORTANT FOR CROPS?

- It is a key constituent of amino acids, enzymes, genetic materials, phyto-synthetic pigments, and energy transfer compounds in a plant.
- Typically, Nitrogen content in a healthy plant is in the range of 1.5-4%.



HOW DOES THE PLANT ABSORB NANO UREA?

- When sprayed on the leaves, Nano urea easily enters through stomata, and other openings and is assimilated by the plant cells.
- It is easily distributed through phloem from source to sink, inside the plants.
- Unutilised nitrogen is stored in the plant vacuole and is slowly released for proper growth and development of plant.



IT IS A GIFT FROM AN YOUNG SCIENTIST!

- Ramesh Raliya, from Rajasthan's Jodhpur district, is the man behind this invention.
- He began working on nano urea in 2009, during his Ph.D at the ICAR.
- He completed its development for use at industrial scale at Washington University, Saint Louis, USA.
- In 2019, Raliya shifted base to India, and joined IFFCO, and set up a nano technology R&D Centre at Gandhinagar.



HOW IS NANO UREA BENEFICIAL?

- Foliar application of nano urea at critical crop growth stages of a plant effectively fulfills its nitrogen requirement and leads to higher crop productivity, and quality in comparison to conventional urea.
- The uptake efficiency increases by 80%. Hence, it requires in lesser measure compared to the conventional urea fertilizer.
- Nano Urea does not involve any government subsidy.

It increases the crop productivity by 8%, and reduces the requirement of conventional urea by 50%. Moreover, it improves yield, biomass and nutritional quality of the produce.

PRADHANMANTRI BHARTIYA JANURVARAK PARIYOJNA (PMBJP)



- Under PMBJP, all fertilisers - Urea, DAP, MoP and Complex (combination of NPK) - will be sold under single “Bharat” brand.
- To explain further, IFFCO's urea will be called "Bharat Urea" so also IPL's MoP become "Bharat MoP".
- The Government has shared the design of the bag, where $\frac{2}{3}$ rd of the space will be used for the brand and logo, while the companies can print their name and details in the balance $\frac{1}{3}$ rd area.
- The logo will also mention Pradhanmantri Bhartiya Janurvarak Pariyojna in Hindi.

1

CENTRE'S CONTENTION IS TO AVOID CROSS- COUNTRY TRANSPORTATION COSTS

The Centre contends that as a major portion of the production cost for fertilisers is met by the exchequer, there's no point in individual fertiliser makers incurring freight costs on shipping their products cross-country, just because farmers prefer one brand over another.



CRITICS SAY THAT THIS MAKES THE PRIVATE PLAYERS LOSE INTEREST IN FERTILISER BUSINESS!

- This argument is weak because freight costs, by the Centre's own admission, account for just Rs. 6,000 crore of the overall subsidy bill of Rs. 2.25 lakh crore.
- While the savings on freight will be small, the bar on branding could prompt fertiliser makers to suspend all marketing, advertising and extension activities that they've so far used to build farmer connect and market share in this commoditised business.
- Private players would also have little incentive to experiment with customised products or new nutrient combinations.
- If these constraints prompt them to reduce their presence in this sector over time, the country may be required to rely on public sector producers, who may not be efficient or cost effective.

2

CENTRE'S CONTENTION IS FERTILISER INVOLVES HUGE SUBSIDY BY THE CENTRE!

- The Centre seems to be keen on making farmers aware that it is footing a hefty bill to make nutrients available to them at ultra-cheap prices.
- For example, for urea, farmer pays just only 11%, whereas the Centre bears 89%, as per the Government. In case of DAP, the Government bears around 65% of the cost and farmers pay just 35%. For MoP, the government subsidy comes to 30%, whereas farmers buy at 70% of the actual cost price.

CRITICS SAY THAT IT IS TO GAIN POLITICAL MILEAGE WITH TAX PAYERS MONEY!

- While it's true that the subsidy component on fertilisers has been shooting through the roof lately, the main reason for this is the Centre's own unwillingness to hike selling prices even marginally to keep up with inflation.
- Branding this scheme with the acronym 'PMBJP' smacks of an attempt to gain political mileage out of an expense, that is funded by taxpayer money.



IT IS HIGHLY RETROGRADE ... SAYS THE EXPERTS!



- The new measures add an additional layer of control to an already over-controlled industry, where selling prices and distribution are dictated by the government.
- The bar on branding will effectively deprive fertiliser makers of the ability to differentiate their products and retain market shares, serving as a deterrent to private presence in the sector.
- A throwback to the licence era - the scheme dashes hopes of reforms that could have set this critical sector on the path to commercial viability and self-sufficiency!

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